

Geometrical optics

→ Ray optics simulator:

<https://phydemo.app/ray-optics/simulator/>

Objectives

→ learn ray optics principles using a simulator to understand the building blocks of fluorescence/confocal microscopy.

Part 1. Foundations: Light and Lenses

Part 2. Diverging Optics and Imaging

Part 3. Numerical aperture and optical assemblies

Part 4. Imperfections of Lenses

Part 5. Building Toward Microscopy

Report – What is expected

Concept: Define the physical concept(s) illustrated by the simulation. Define the relevant parameters and symbols (e.g., focal length, numerical aperture, chromatic aberration).

Simulation setup: Briefly describe the optical configuration and indicate the relevant parameters.

Illustration: Include a numbered screenshot of the relevant optical layout. Annotate the important elements (source, lens, optical axis, focal point/image plane, detector, etc., when relevant).

Observation and Interpretation: Describe what changes or what you observe in the simulation. Explain why you observe it using the physical concept and, when relevant, the corresponding equation.

Equations: When a quantitative relationship is relevant, identify the appropriate equation, define every symbol, and explain how it relates to the simulation.

Scientific presentation:

- Every figure must be cited in the main text.
- Place figures close to their first citation.
- A figure legend describes the figure; it does not replace the explanation in the main text.
- Define technical terms and parameters when first introduced.

Bonus:

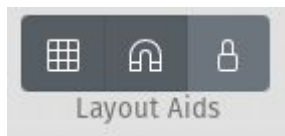
- Point out where ray (geometrical) optics is a simplification and when wave optics (diffraction, interference) becomes essential

Practical tips

- Start using ideal lenses

- Click on GRID

- Click on Snap To Grid



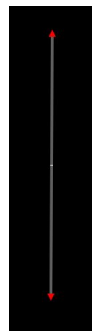
- You can insert a detector anywhere and tick its “Irradiance Map” to see the intensity profile

- Tune “ray density” -/+

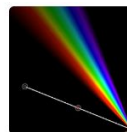
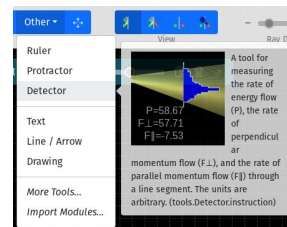


- To play with multiple wavelengths:

- In “Other” → import modules → ContSpectrum
- In Settings → Simulate colors



***Ideal lens:** thin-lens model without aberrations*



Continuous spectrum light source ([ContSpectrum](#))

[Import](#)

Yi-Ting Tu

A light source with a uniform continuous spectrum discretized with a given constant step. Only works when “Simulate Colors” is on. [Specification](#)

Part 1. Foundations: Light and Lenses

1. Light sources

- *Simulation:* Laser (collimated), LED (diverging), isotropic point source (simplified model of fluorescence emission).
- *Goal:* Define collimated, diverging and isotropic. Compare the three sources and relate them to excitation and fluorescence emission.

2. Single convex lens with collimated light

- *Simulation:* Collimated laser through convex lens.
- *Goal:* Identify the optical axis and focal point. Explain what the lens does to parallel rays.

3. Single convex lens with divergent light

- *Simulation:* Move a divergent point source in front of an ideal converging lens.
- *Goal:* Describe how changing the source position changes the position of the image. Distinguish the lens focal point from the image point.

Part 2. Diverging Optics and Imaging

1. Concave lens with collimated beam

- *Simulation:* Collimated beam through an ideal diverging lens ($f < 0$).
- *Goal:* Show the diverging rays and identify the virtual focal point. Explain the meaning of a negative focal length.

2. Image formation with lenses

- *Simulation:* Represent an extended object using several point sources. Form its image through an ideal converging lens. Move the object and locate the corresponding image plane. Keep a detector at a fixed position to illustrate defocus.
- *Goal:* Explain how rays originating from each object point form the corresponding image point. Distinguish real and virtual images, and investigate the effect of the detector position on image formation. Identify the relationship between object distance, image distance and focal length.

Part 3. Numerical aperture and optical assemblies

1. Numerical Aperture (NA)

- *Simulation:* Keep the point source-lens distance fixed and vary the effective lens aperture.
- *Goal:* Define NA, give its mathematical expression and explain how changing the collection angle affects the amount of collected light.

2. Beam expanders (telescope configuration)

- *Simulation:* Two convex lenses with different focal lengths.
- *Goal:* Show that the input and output beams are collimated and identify the relationship between focal lengths and beam diameter magnification.

Part 4. Imperfections of Lenses

1. Plano-convex lens orientation

- *Simulation:* Flip plano-convex lens relative to incoming collimated beam.
- *Goal:* Compare both lens orientations and discuss their effect on image quality.

2. Chromatic aberrations

- *Simulation:* Polychromatic (multi-wavelength) light through a real spherical converging lens.
- *Goal:* Compare the behaviour of different wavelengths and discuss the consequences for imaging; link to need for achromats/apochromats.

Recall: Refraction at a real optical interface is governed by the Snell-Descartes law.

Part 5. Building Toward Microscopy

1. Epifluorescence setup (basic)

- *Simulation:*
 - Build the fluorescence detection path from the sample to the camera: sample → objective → dichroic mirror → emission filter → tube lens → camera.
 - Then add the excitation path: excitation source → excitation filter → dichroic mirror → objective → sample.
- *Goal:* Identify the sample, objective, dichroic mirror, excitation/emission filters, tube lens and detector. Describe separately the excitation and emission paths. Identify where light is focused or collimated and explain the role of the dichroic mirror and filters.

Part 5. Building Toward Microscopy

2. Confocal setup (conceptual)

- *Simulation:* Compare fluorescence originating from an in-focus and an out-of-focus point. Place a pinhole in the corresponding image plane before the detector.
- *Goal:* Determine the role of the pinhole. How should the excitation light be arranged? How can a complete image be formed?